Mozes Jacobs

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Education

Harvard University — Cambridge, MA
Ph.D. in Computer Science | Advisor: Demba Ba | Kempner Graduate Fellow — 2023 – **May 2027 (Expected)**
Mechanistic Interpretability, Representation Learning, Robotics

University of Washington — Seattle, WA
B.S. in Computer Science | Summa Cum Laude | GPA: 3.99 / 4.00 — 2018 – 2022

Research Experience

Goodfire AI | *Research Fellow* — May – Sep 2026

- Led a mechanistic study of VLA policies, using controlled sweeps, probes, and activation interventions to map the geometry of 7 task variables across simulated and real robots; steered behavior along this geometry.
- Developed block-sparse featurizers that recover multi-dimensional concept manifolds in vision, diffusion, and robotics models, enabling fine-grained steering and data attribution isolating <1.4% of training tokens behind a behavior.
- Localized a narrow action-conditioning interface in 3 frontier VLAs ($\pi_{0.5}$, MolmoAct2, MolmoBot) via activation patching; restricting conditioning to 3/36 layers cut inference FLOPs by up to 33% at matched success.

Kempner Institute | **Harvard University** | *Graduate Research Fellow* — 2023 – Present

- Proposed the Block-Recurrent Hypothesis for ViTs; trained 2-block recurrent surrogates recovering 96% of DINOv2's ImageNet-1k probe accuracy and used them to reveal attractor basins and low-rank late-layer dynamics.
- Disentangled attention sinks into two distinct algorithms (nop and broadcast), derived diagnostics for each, and showed that combining gating with registers yields complementary segmentation gains.

Allen Institute | **AI Institute in Dynamic Systems** | *Shanahan Foundation Predoctoral Fellow* — 2022 – 2023

- Built HyperSINDy, a deep generative framework that discovers sparse stochastic governing equations from data with uncertainty quantification. *(Advisors: R. Raut, J.N. Kutz, S. Brunton)*

Publications

*equal contribution ‡core contributor

[1] **A Phenomenology of Neural Geometry in Vision-Language-Action Models** — **arXiv 2026 · Under Review**
Jacobs M.*, Swann A.*, Papillon M.*, et al.

[2] **Structuring Sparsity: Block-Sparse Featurizers Capture Visual Concept Manifolds** — **arXiv 2026 · Under Review**
Fel T.*, Kowal M.*, **Jacobs M.***, Hazra D.*, Bhalla U.*, et al.

[3] **Multi-Dimensional Featurizers Yield Fine-Grained Steering and Data Shaping in Robot Foundation Models** — **arXiv 2026 · Under Review**
Swann A.‡, **Jacobs M.‡**, Papillon M.‡, et al.

[4] **Causal Analysis of Robot Foundation Models Enables Efficiency Gains** — **arXiv 2026 · Under Review**
Swann A.‡, Papillon M.‡, **Jacobs M.‡**, et al.

[5] **A Unifying View of Attention Sinks: From Mechanisms to Architectural Interventions** — **arXiv 2026 · Under Review**
Fesser L.*, **Jacobs M.***, Fel T.*, Keller TA., Kakade S.

[6] **Block-Recurrent Dynamics in ViTs** — **ICLR 2026**
Jacobs M.*, Fel T.*, Hakim R.*, Brondetta A., Ba D., Keller TA.

[7] **Traveling Waves Integrate Spatial Information Through Time** — **CCN 2025 (Oral)**
Jacobs M., Budzinski RC., Muller L., Ba D., Keller TA.

[8] **HyperSINDy: Deep Generative Modeling of Nonlinear Stochastic Governing Equations** — **arXiv 2023**
Jacobs M., Brunton BW., Brunton SL., Kutz JN., Raut RV.

Talks

[1] **Traveling Waves Integrate Spatial Information Through Time**
Neural Computations: Dynamics Across Space, Time, and Task — **CCN 2025 (Aug 2025)**

[2] **Can Your Neurons Hear the Shape of an Object?**
Frontiers in NeuroAI Symposium — **Kempner Institute (June 2025)**

Teaching Experience

Harvard University | *Teaching Fellow, APMTH 231: Decision Theory (Prof. Demba Ba)* — Spring 2026
Harvard University | *Teaching Fellow, CS 2790R: Human-Computer Interaction (Prof. Elena Glassman)* — Fall 2025

Awards & Fellowships

Kempner Institute Graduate Fellowship — 2023 – Present
Shanahan Foundation Predoctoral Fellowship — 2022
UW Scholarships: Burkhardt Family Endowed (2020, 2021), Gary A. Kildall Endowed (2019), Boeing (2018) — 2018 – 2021

Technical Skills

Languages & Frameworks: Python, PyTorch, Hugging Face Transformers, NumPy, Java
Interpretability: SAEs, block-sparse featurizers, activation patching, causal interventions, probing, steering, data attribution
Foundation Models: Training, fine-tuning & distillation of ViTs (DINOv2/v3), VLAs ($\pi_{0.5}$, MolmoAct2, MolmoBot), SDXL
Robotics: VLA policies, robot manipulation, closed-loop policy evaluation, sim & real-world robot data, real-robot deployment
Infrastructure: Multi-GPU distributed training, SLURM/HPC clusters, Linux, Git, Weights & Biases